

Task 2: Conditional SELECT Queries

- (a) List all employees whose age is between 25 and 40, ordered by last name ascending.

```
SELECT * FROM EMPLOYEE
```

```
WHERE AGE BETWEEN 25 AND 40
```

```
ORDER BY LNAME ASC;
```

```
SELECT * FROM EMPLOYEE
WHERE AGE BETWEEN 25 AND 40
ORDER BY LNAME ASC;
```

EMP_ID	FNAME	LNAME	GENDER	AGE	CONTACT_ADD	EMP_EMAIL	EMP_PASS	JOB_ID
1	Ali	Ahmed	M	28	Muscat	ali@ems.com	123	1
2	Omar	Ali	M	30	Muscat	omar@ems.com	123	5
3	Mona	Ali	F	31	Muscat	mona@ems.com	123	2
4	Nasser	Hamad	M	35	Sohar	nasser@ems.com	123	2
5	Fatma	Hassan	F	27	Sohar	fatma@ems.com	123	4
6	Aisha	Mohammed	F	29	Muscat	aisha@ems.com	123	3
7	Salim	Nasser	M	33	Barka	salim@ems.com	123	4
8	Noor	Said	F	25	Sur	noor@ems.com	123	5
9	Khalid	Said	M	40	Nizwa	khalid@ems.com	123	3
10	Sara	Salim	F	26	Barka	sara@ems.com	123	1

- (b) Retrieve all payroll records where total_amount exceeds 5000, showing employee name and department.

```
SELECT E.FNAME, E.LNAME, J.JOB_DEPT, P.TOTAL_AMOUNT FROM PAYROLL P
JOIN EMPLOYEE E ON P.EMP_ID = E.EMP_ID JOIN JOB_DEPARTMENT J
ON P.JOB_ID = J.JOB_ID WHERE P.TOTAL_AMOUNT > 5000;
```

```
SELECT E.FNAME, E.LNAME, J.JOB_DEPT, P.TOTAL_AMOUNT FROM PAYROLL P
JOIN EMPLOYEE E ON P.EMP_ID = E.EMP_ID JOIN JOB_DEPARTMENT J
ON P.JOB_ID = J.JOB_ID WHERE P.TOTAL_AMOUNT > 5000;
```

FNAME	LNAME	JOB_DEPT	TOTAL_AMOUNT
1 Ali	Ahmed	Engineering	6500
2 Ali	Ahmed	Engineering	6500
3 Nasser	Hamad	Finance	7700
4 Nasser	Hamad	Finance	7700
5 Mona	Ali	IT	5900
6 Khalid	Said	Marketing	5350

- (c) Find all employees who have taken leave with reason containing the word 'sick' (case-insensitive).

```
SELECT E.FNAME,E.LNAME,L.REASON FROM EMPLOYEE E
```

```
JOIN LEAVE_RECORD L ON E.EMP_ID = L.EMP_ID
```

```
WHERE LOWER(L.REASON) LIKE '%sick%';
```

```
SELECT E.FNAME,E.LNAME,L.REASON FROM EMPLOYEE E
JOIN LEAVE_RECORD L ON E.EMP_ID = L.EMP_ID
WHERE LOWER(L.REASON) LIKE '%sick%';
```

Query Result		
SQL All Rows Fetched: 2 in 0.033 seconds		
FNAME	LNAME	REASON
1 Ali	Ahmed	Sick Leave
2 Khalid	Said	Sick Leave

- (d) List all departments that have no employees assigned. (Use outer join or NOT EXISTS.)

```
SELECT J.* FROM JOB_DEPARTMENT J LEFT JOIN EMPLOYEE E
```

```
ON J.JOB_ID = E.JOB_ID WHERE E.EMP_ID IS NULL;
```

```
SELECT J.* FROM JOB_DEPARTMENT J LEFT JOIN EMPLOYEE E
ON J.JOB_ID = E.JOB_ID WHERE E.EMP_ID IS NULL;
```

Query Result				
SQL All Rows Fetched: 0 in 0.011 seconds				
JOB_ID	JOB_DEPT	NAME	DESCRIPT...	SALARY_...